Supplement, appendix and contribution statement

Everything here supports the main tab; none of it changes a registered result. Post-hoc and synthetic analyses are labelled as such.

S1. Model selection on validation data (F0 → V)

Metric: pinball loss at q = 0.90 on the unshifted integer promise max{1, ⌈ŷ⌉}. A linear step was adopted only if it lowered the loss by at least 1%. Candidates within 1.001 × the global minimum counted as tied; inside a tie the simpler setting won.

Candidate	V loss	Change vs current best	Decision
C0 registered linear (log1p target, numeric weekday, no penalty)	1.28856	—	start
C1 weekday as a category	1.29170	−0.24%	rejected
C2 raw-days target	1.28932	−0.06%	rejected
C3 L1 penalty, best of 3 alphas (0.0001)	1.28713	+0.11%	rejected (< 1%)
D3 HGB, depth 2, 400 iterations (best of 8)	1.26266	—	selected; tie set D3, D4, D7

Post-hoc intervals on V. A week-block bootstrap on V (14 weeks, 2,000 draws) gives an improvement over C0 of 0.11% (−0.002% to 0.245%) for the C3 pick and 2.01% (1.11% to 2.85%) for D3. The first interval shows no improvement reaching the pre-set 1% bar, so the original choice stands. Both candidates were picked on this same V, and the intervals do not include selection or refitting uncertainty; therefore D3′s figure is optimistic and was only a reason to evaluate it.

S2. A small simulation of two calibration failures (synthetic data)

Question: does a calibration shift fail because the delivery distribution changes over time, or because at the freeze only the fast, already-delivered orders can be used? Delivery days follow log Y = 2.2 + 0.5x + 0.45ε, rounded up. The predictor is the conditional median, and the calibration code is the project's. Each of 1,000 replications draws 2,000 calibration and 5,000 evaluation orders (seed 20261021). All scenarios share the evaluation sample. For the time-change scenarios, the continuous delivery time is rescaled before rounding.

Scenario minus the stable, complete-label scenario	Coverage change, pts (mean ± MC s.e.)	Middle 90% of replications	Promised days change
Only delivered-by-freeze labels (90-day window)	−1.68 ± 0.015	−2.48 to −0.94	−0.99
Random subset of the same size (diagnostic)	+0.02 ± 0.009	−0.42 to +0.46	+0.02
Later deliveries 15% slower	−5.77 ± 0.014	−6.50 to −5.08	0
Later deliveries 15% faster	+4.26 ± 0.012	+3.66 to +4.90	0

The two mechanisms leave different traces. Selective labels shrink the shift itself, so promised days and coverage fall together; a random subset of the same size does not. A later change in delivery times leaves the promise unchanged and moves only coverage. The "middle 90%" column is the spread of replications, not a confidence interval for the mean. A first, unpaired run (seed 20261020) gave the same picture. This explains mechanisms; it does not show which one, if either, acted in the Olist data.

A. Reproducibility and checks

Item	Value
Pre-registered configuration	config/date_model_e1.yaml v3, SHA-256 ae6a4a46…20152
Freeze record	FREEZE.json, SHA-256 3b62af1a…328dd; 9 runs; the final E1 scoring run came after it (E had been examined in Stage 7)

Item	Value
Automated tests	317 pass, including synthetic end-to-end tests showing that scoring stops before any E output if the freeze, a model file, the sample, the calibration shifts or (after the post-hoc hardening) any input content changed, or if the post-hoc addendum is malformed or bound to a different freeze
Independent recomputation	every reported number recomputed from the saved per-order files with plain NumPy / pandas: 43 / 43 for E results, 43 / 43 for supplementary results
Post-hoc hardening	content hashes of all inputs, added after scoring as a labelled addendum; the original freeze record is unchanged
Environment	Python 3.13.16; numpy 2.5.3, pandas 3.0.5, scikit-learn 1.9.1, scipy 1.18.1
Data	public Olist e-commerce dataset (Kaggle); raw files and order-level marts are not redistributed
Not done	a full rebuild from the raw CSVs by an independent reviewer; a second evaluation period

B. Contribution statement (draft — the author must check every line)

Part	Who	Author to confirm
Reframing the question from late-order prediction to promise setting	Author, through the project and discussions	[ ]
Experiment scope and constraints (three model families, no tuning on E, staged checkpoints, disclosure rule)	Author	[ ]

Part	Who	Author to confirm
Pre-registration text, code for selection, freeze and scoring, tests, simulation	Drafted and implemented with an AI assistant (Claude), revised over several review rounds	[ ]
Reviews that found design and code issues (comparison confound, freeze gaps, wording)	External review rounds [author: name the source]	[ ]
Libraries	scikit-learn (QuantileRegressor, HistGradientBoostingRegressor, mean_pinball_loss definition), NumPy, pandas, SciPy	—
Results I recomputed or re-derived myself	[author: list, e.g. the integer-day derivation, the V check, one E comparison]	[ ]

The lines above should be filled from the actual records, not from memory: docs/E1_DECISIONS.md logs every change to the plan with its time, its reason and whether it came from the author's go-ahead or from a review; the review reports and the evidence packages are kept with it. The author should be able to explain, without notes: the integer-day derivation; why the calibration shift uses the (n + 1)q-th residual; why C* and B3 are the same model; and why a lower pinball loss gave longer HGB dates.